

1. THEORETICAL FOUNDATION

Problem: dam break is a two-phase (water + air) flow. A water column is released instantly and collapses under gravity, so the region occupied by water is unknown and changes every step.

NAVIER-STOKES (MOMENTUM, MIXTURE)

$$\rho [\partial u / \partial t + (u \cdot \nabla) u] = -\nabla p + \nabla \cdot [\mu (\nabla u + \nabla u^T)] + \rho g$$

CONTINUITY (MASS)

$$\nabla \cdot u = 0$$

MIXTURE PROPERTIES (LINEAR BLEND OF A)

$$\rho = \alpha \cdot \rho_{\text{water}} + (1 - \alpha) \cdot \rho_{\text{air}}$$

$$\mu = \alpha \cdot \mu_{\text{water}} + (1 - \alpha) \cdot \mu_{\text{air}}$$

VOF method: a fixed grid spans both fluids. α = water volume / cell volume: $\alpha = 1$ water, $\alpha = 0$ air, $0 < \alpha < 1$ interface cell. ρ and μ are recomputed from α every step.

2. ASSUMPTIONS AND WALL CONDITIONS

WHAT STARTS THE MOTION

In the dam break, gravity is the only thing that starts the motion. Nothing pushes the water; the column just collapses under its own weight.

CONTINUUM ASSUMPTION

Water and air are treated as smooth, continuous fluids, not as individual molecules. Every cell then has a density, velocity and pressure.

CONDITION	MEANING
No-penetration	Fluid cannot cross the wall. Always true for solid walls.
No-slip	Fluid touching the wall is stuck to it (zero speed there). Realistic for real walls.
Free-slip	Fluid slides along the wall with no friction. An idealisation.
Open boundary	Fluid and air can pass freely, like the open top of a tank.

3. INTERFACE TRANSPORT

A TRANSPORT: NO DIFFUSION, NO SOURCE

$$\partial\alpha/\partial t + u \cdot \nabla\alpha = 0$$

equivalent to $\partial\alpha/\partial t + \nabla \cdot (\alpha u) = 0$ because $\nabla \cdot u = 0$ (conservative form)

SCHEME	WHAT IT DOES	TRADE-OFF
Plain upwind	α treated like any other scalar	Easy, but interface smears
Donor-acceptor (USED)	Blends upwind/downwind by interface orientation and donor fullness	Sharp without geometry; needs CFL < 1
PLIC	Straight-line interface per cell, exact geometric fluxes	Sharpest; many edge cases
Van Leer	Flux limiter for momentum, not for α	High order when smooth, bounded at jumps

4. TIME STEP: CHORIN PROJECTION

1. PREDICTOR (NO PRESSURE)

$$u^* = u + \Delta t \cdot [-(u \cdot \nabla) u + (1/\rho) \nabla \cdot (\mu \nabla u) + g]$$

2. PRESSURE POISSON

$$\begin{aligned} \nabla \cdot ((1/\rho) \nabla p) \\ = (1/\Delta t) \nabla \cdot u^* \end{aligned}$$

3. PROJECTION

$$\begin{aligned} u(n+1) &= u^* - (\Delta t/\rho) \nabla p \\ \text{so that } \nabla \cdot u(n+1) &= 0 \end{aligned}$$

4. Advect α with the corrected velocity (donor-acceptor), then clip α to $[0, 1]$.

5. Recompute ρ and μ from α ; repeat with a CFL-limited Δt .

Why split? u^* is where the water wants to go from gravity, momentum and viscosity alone. The Poisson solve finds the pressure that removes the divergence of u^* ; the projection applies it.

5. DENSITY RATIO AND PRESSURE SOLVE

FLUID	DENSITY (ρ)	KINEMATIC VISCOSITY (ν)	RATIO
Water	998.0 kg/m ³	1.00exp(-6) m ² /s	~832× (ρ)
Air	1.20 kg/m ³	1.48exp(-5) m ² /s	

WHY IT IS ILL-CONDITIONED

The coefficient $1/\rho$ jumps ~832× across a single interface cell, so the Poisson matrix entries span about three orders of magnitude and plain CG converges slowly.

FIX: IC(0)-PRECONDITIONED CG

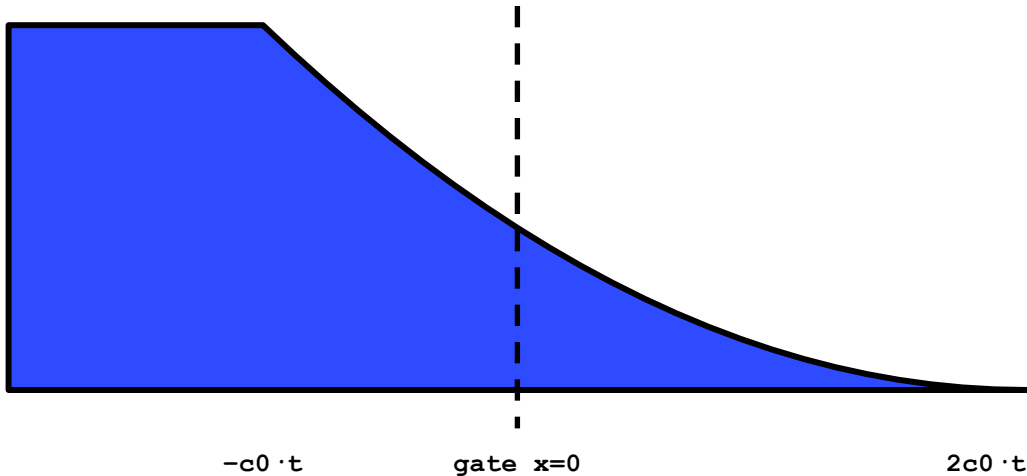
2450 → 112 iterations

18.5 → 2.1 ms per step; scaling $O(N^{1.5}) \rightarrow O(N^{1.2})$

Matrix assembly: the 5-point Laplacian must treat fluid and empty cells carefully to stay symmetric positive definite.

6. DAM-BREAK THEORY: RITTER

WATER AT REST ($h = H$)



SHALLOW WATER (SAINT-VENANT)

$$h_t + (h \cdot u)_x = 0$$

$$u_t + u \cdot u_x + g \cdot h_x = 0$$

RITTER SOLUTION, $c_0 = \sqrt{gH}$

$$u = \frac{2}{3} (c_0 + x/t)$$

$$h = \frac{(2 \cdot c_0 - x/t)^2}{9g}$$

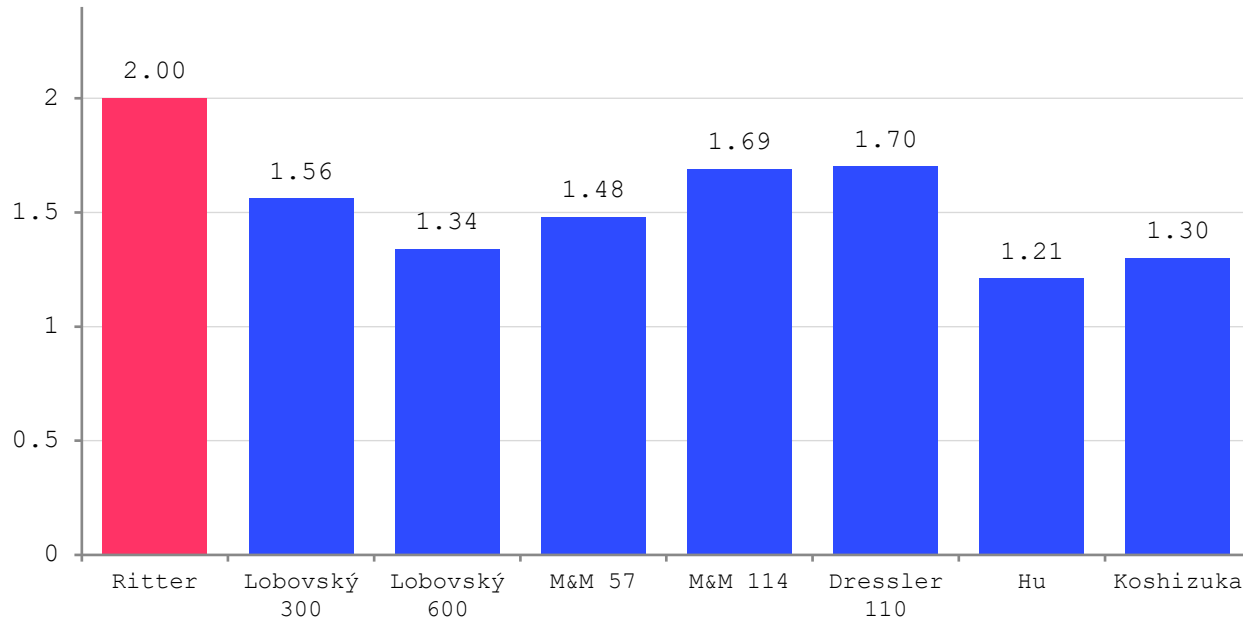
for $-c_0 \cdot t \leq x \leq 2 \cdot c_0 \cdot t$

WHAT IT PREDICTS (VALID FOR $T < L_0/c_0$, BEFORE REFLECTION OFF THE BACK WALL)

Front speed $2\sqrt{gH}$ with zero depth. At the gate: $h = 4H/9$, $u = c$ ($Fr = 1$), constant discharge $q = (8/27) \cdot \sqrt{g} \cdot H^{1.5}$.

7. REALITY CHECK: FRICTION AND SCALING

AVERAGE FRONT SPEED $v/\sqrt{gH}$, $t^* > 1$



FRICTION (DRESSLER, WHITHAM)

Blunt front, vertical tangent, finite depth. Slower than $2\sqrt{gH}$.

LAB SCALE, $H = 300$ MM

$Re = 3.8e6$, $We = 1.64e5$, $Fr = 1$

Large We : surface tension negligible

EARLY TIMES, $T^* < 1$

Not hydrostatic; gate removal (3.5–4.5 m/s) matters.

Source: Lobovský et al., Table 1. M&M = Martin & Moyce; numbers after names are H in mm.

8. VALIDATION

SOURCE	LENGTH	TIME
Martin & Moyce	initial width a	$T = t\sqrt{(g/a)}$
Lobovský	depth H	$t^* = t\sqrt{(g/H)}$
This solver	L_0	$T^* = t\sqrt{(g/L_0)}$

EARLY-TIME CHECK (RITTER)

$$X^* = 1 + 2 \cdot T^*$$

front starts at $x/L_0 = 1$, then moves at $2\sqrt{(gH)}$; square column

Match the axis convention of the data you overlay before comparing curves.

ACCEPTANCE CHECKS

- $\int \alpha \, dV$ stays within a stated tolerance (live readout)
- Front $X^*(T^*)$ vs Martin & Moyce data
- Front stays below Ritter: $1 + 2T^*$
- Late front speed about $1.1\text{--}1.75 \sqrt{(gH)}$
- $\nabla \cdot \mathbf{u} \approx 0$ after every projection
- Error shrinks under grid and Δt refinement

9. IMPACT PRESSURE AT A WALL

MEDIAN PEAK, 3 MM ABOVE BED

$$\approx 3 \times \rho g H$$

97.5th percentile $\approx 4.5 \times$

P / PV^2 , MEASURED

1.25

vs 0.5 for a steady impinging
jet

REPEATED RUNS PER FILL HEIGHT

100

peak pressure is a random
variable

Time scales: rise 1.5–4.5 ms, decay about 10× longer. Impulse $\int P \, dt \approx \frac{1}{2} \times \text{peak} \times \text{impact time}$ (within about 25%).

Scaling: lowest-sensor peak $\propto H$ (Froude: $P \sim \rho g H$); higher sensors are not linear in H .

For your solver: incompressible VOF gives a sharp, grid-dependent spike. Compare arrival time and impulse with the median and 95% band, not one peak.

10. SCOPE FOR IMPROVEMENT

HIGHER-ORDER INTERFACE TRACKING

Replace donor-acceptor with PLIC for a continuous, sharp geometric interface reconstruction.

HARDWARE ACCELERATION

Port grid sweeps and the Poisson solver to WebGPU compute shaders; scale to millions of cells and 3D.

PHYSICS GAPS

Air compressibility and entrapment at impact, wet-bed jets, and turbulence at $Re \sim 1e6$ on an under-resolved grid.

3D EFFECTS

The lab flow stops being 2D at $H = 600$ mm, which a 2D solver cannot show.